CSE331
AUTOMATA AND COMPUTABILITY

Designing Turing Machine

Spring 2026

Turing Machine: High Level description

High Level description of TM to decide $L = \{0^{2^n} : n \geq 0\}$

Algorithm Powers of 2(w) ▷ check if $|w| = 2^n$

```
while true do
    if |w| = 1 then
        return Accept
    else if |w| is odd then
        return Reject
    else
        delete half of the 0's in w
```

In every iteration, the number of 0's on the tape is halved. The string is accepted if and only if the number of 0's is a power of 2.

Turing Machine: Medium Level description

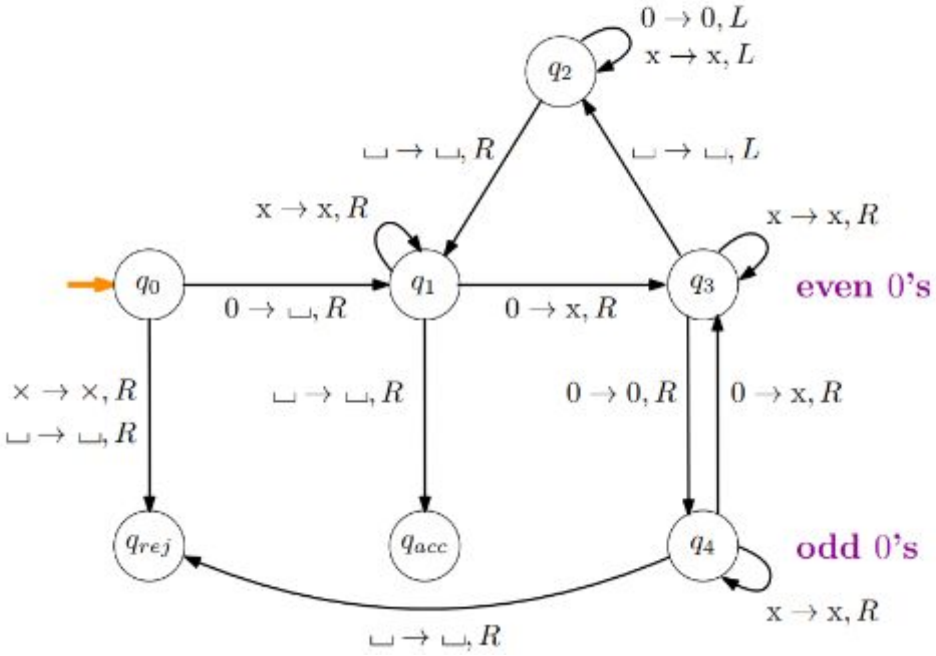
Medium Level description of TM to decide $L = \{0^{2^n} : n \geq 0\}$

1. Move the head from left to right, cross out every other 0
   a. If in step 1 there is only one 0, **accept**
   b. If in step 1 there is an odd (> 1) number of 0's **reject**
2. Return the head back to the left end of the tape
3. Go back to step 1

Again, in every iteration, the number of 0's on the tape is halved. The string is accepted if and only if the number of 0's is a power of 2.

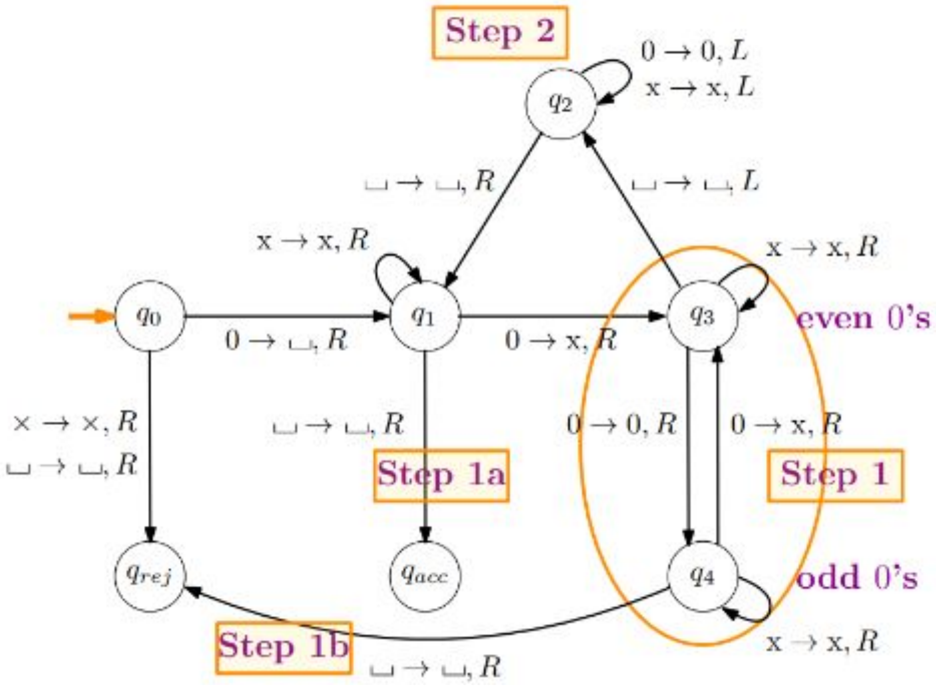
Turing Machine: Low Level description

Low Level description of TM to decide $L = \{0^{2^n} : n \geq 0\}$



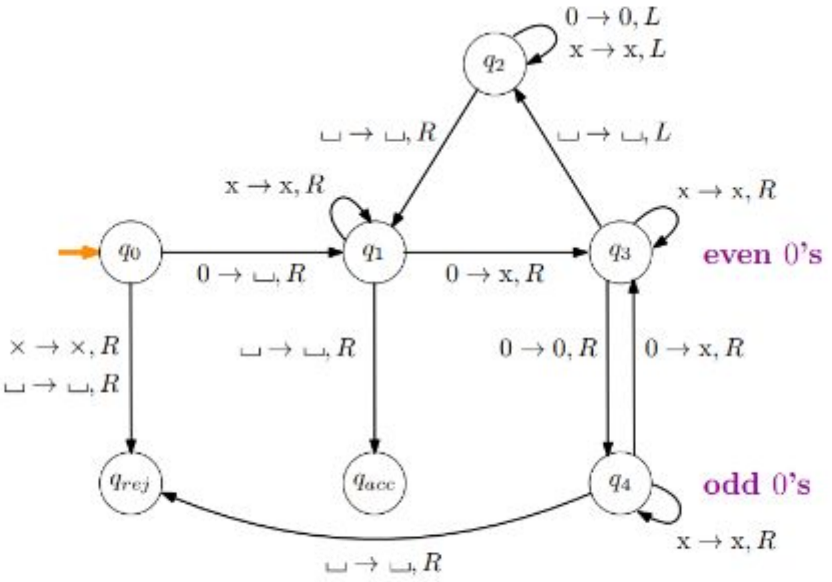
Turing Machine: Low Level description

Low Level description of TM to decide $L = \{0^{2^n} : n \geq 0\}$



Turing Machine: Low Level description

Low Level description of TM to decide $L = \{0^{2^n} : n \geq 0\}$



$q_0 0000\sqcup$
$\sqcup q_1 000\sqcup$
$\sqcup x q_3 00\sqcup$
$\sqcup x0 q_4 0\sqcup$
$\sqcup x0x q_3 \sqcup$
$\sqcup x0 q_2 x\sqcup$
$\sqcup x q_2 0x\sqcup$
$\sqcup q_2 x0x\sqcup$
$q_2 \sqcup x0x\sqcup$
$\sqcup q_1 x0x\sqcup$
$\sqcup x q_1 0x\sqcup$
$\sqcup xx q_1 x\sqcup$

$\sqcup xx q_3 x\sqcup$
$\sqcup xxx q_3 \sqcup$
$\sqcup xx q_2 x\sqcup$
$\sqcup x q_2 xx\sqcup$
$\sqcup q_2 xxx\sqcup$
$q_2 \sqcup xxx\sqcup$
$\sqcup q_1 xxx\sqcup$
$\sqcup x q_1 xx\sqcup$
$\sqcup xx q_1 x\sqcup$
$\sqcup xxx q_1 \sqcup$
$\sqcup xxx\sqcup q_{acc}$

Run the TM on ϵ, 0, 00, 000